Generalized Phase Shift Pilots for Massive MIMO-OFDM Channel Estimation

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***Abstract*—We investigate a non-orthogonal pilot design for massive multiple-input multiple-output (MIMO) channel estimation with orthogonal frequency division multiplexing (OFDM) modulation. With beam based channel model, we formulate a signal model for channel acquisition with a comb-type pilot structure and derive its simplified equivalent form. Then, we propose generalized phase shift pilots (GPSPs) and reveal their relevant properties. We prove that the inter-user interference with the proposed pilot is negligible when the weight sequence length is large enough, validating its feasibility, especially when the statistical channel state information (CSI) is unavailable. By leveraging the properties of the GPSPs, we provide an efficient implementation for channel estimation with the Gaussian generalized approximate message passing sparse Bayesian learning (GGAMP-SBL) algorithm, where the number of complex multiplications is significantly reduced. Simulation results indicate that the proposed pilot enables low-complexity channel estimation while satisfactory performance is ensured.**

***Index Terms*—Massive MIMO-OFDM, beam based channel model, generalized phase shift pilots, channel estimation**

I. Introduction

Massive multiple-input multiple-output orthogonal frequency division multiplexing (MIMO-OFDM) [1] is a fundamental enabling technology for the fifth generation (5G) mobile communication, and will play a key role for the sixth generation (6G) mobile communication with increased system configuration in antenna and user terminal (UT) numbers. In massive MIMO systems, a large number of antennas are deployed at the base station (BS) to simultaneously serve a number of UTs on the same time and frequency resources. Therefore, massive MIMO can effectively exploit spatial wireless resources and significantly enhance spectral efficiency [2].

Acquiring channel state information (CSI) is essential to achieve the potential system capacity provided by massive MIMO-OFDM. Due to the increasing numbers of antennas, subcarriers and UTs in massive MIMO-OFDM systems, conventional orthogonal pilot methods suffer excessive overhead. Therefore, non-orthogonal pilot design, which allows more UTs to be served on the same time and frequency resources, is necessary for massive MIMO-OFDM channel acquisition and has been widely investigated. The limited angle spread in massive MIMO channels allows to reuse the same pilot for UTs with non-overlapping angle domain channels [3]. Assuming the knowledge of visible regions is known, a two-stage graph partitioning is proposed to design non-orthogonal pilots, which reduces the pilot overhead without compromising sum rate performance [4]. Adjustable phase shift pilots (APSPs) with a pilot phase shift scheduling algorithm can fully utilize wireless resources in spatial and frequency dimensions, which enables serving much more UTs than phase shift orthogonal pilots (PSOPs) with similar performance [5]. Nevertheless, these pilots require the statistical CSI or at least the spatial statistical CSI. Since obtaining statistical CSI is relatively hard before channel estimation [6], the application of these pilots might be limited in the practical systems. Thus, a non-orthogonal pilot that can provide satisfactory performance without the statistical CSI is in great demand.

In this paper, we begin by formulating the signal model for channel acquisition under comb-type pilot structure with beam-based channel model. Subsequently, we propose generalized phase shift pilots (GPSPs) and reveal their relevant properties. We show that the inter-user interference associated with GPSPs can be ignored when the length of weight sequences is large enough. Further, we present a low complexity implementation for channel estimation with the Gaussian generalized approximate message passing sparse Bayesian learning (GGAMP-SBL) [7] algorithm by utilizing the properties of GPSPs.

II. System Model

A. System Configuration

We consider a massive MIMO-OFDM system operating in time-division duplex (TDD) mode, where the BS is equipped with a uniform linear array consisting of M antennas, serving U single antenna UTs. The spacing between adjacent antennas is $d = \frac{\lambda_c}{2}$, where λ_c is the wavelength corresponding to the center frequency f_c. In a single OFDM symbol, the number of subcarriers is N_{sc}, the length of cycle prefix (CP) is N_{cp}, the subcarrier interval length is Δf and the sample interval length is $T_s = 1/(N_{sc}\Delta f)$. We assume there are N_v valid

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subcarriers with index started from k_0 used for pilot and data transmission. Due to the reciprocity in TDD systems, we focus on the channel acquisition in UL transmission.

B. Beam Based Channel Model

Assume that the CSI remains unchanged over an OFDM symbol during the UL transmission. The channel between the UT u and the BS in space-frequency (SF) domain can be expressed as [8]

$$\mathbf{H}_u^{\mathrm{SF}} \triangleq \sum_{p=0}^{P_u-1} \gamma_{u,p}\mathbf{v}_{\mathrm{s}}(\Omega_{u,p})\mathbf{v}_{\mathrm{f}}^{\mathrm{T}}(\tau_{u,p}) \in \mathbb{C}^{M\times N_{\mathrm{v}}}, \quad (1)$$

where P_u denotes the number of paths between the UT u and the BS, $\gamma_{u,p}$ and $\tau_{u,p}$ are complex-valued gain and propagation delay at the first antenna of the p-th path between UT u and the BS, respectively, $\Omega_{u,p} \triangleq f_{\mathrm{c}}\Delta\tau \sin\theta_{u,p}^{\mathrm{az}}\cos\theta_{u,p}^{\mathrm{el}} = \frac{1}{2}\sin\theta_{u,p}^{\mathrm{az}}\cos\theta_{u,p}^{\mathrm{el}}$ is the scaled directional cosine with $\theta_{u,p}^{\mathrm{az}}$, $\theta_{u,p}^{\mathrm{el}}$, $\Delta\tau = d/c$ and c representing the azimuth, elevation angles of arrival (AoA), the delay between adjacent antennas and the speed of light, respectively, and $\mathbf{v}_{\mathrm{s}}$, $\mathbf{v}_{\mathrm{f}}$ are steering vectors defined as

$$\mathbf{v}_{\mathrm{s}}(\Omega) \triangleq \left[1, e^{-\bar{\jmath}2\pi\Omega}, \ldots, e^{-\bar{\jmath}2\pi(M-1)\Omega}\right]^{\mathrm{T}}, \quad (2)$$

$$\mathbf{v}_{\mathrm{f}}(\tau) \triangleq \left[e^{-\bar{\jmath}2\pi k_0\Delta f\tau}, e^{-\bar{\jmath}2\pi k_1\Delta f\tau}, \ldots, e^{-\bar{\jmath}2\pi k_{N_{\mathrm{v}}-1}\Delta f\tau}\right]^{\mathrm{T}}. \quad (3)$$

Thus, the element of $\mathbf{H}_u^{\mathrm{SF}}$ can be expressed as $[\mathbf{H}_u^{\mathrm{SF}}]_{i,k} \triangleq \sum_{p=0}^{P_u-1}\gamma_{u,p}e^{-\bar{\jmath}2\pi k\Delta f\tau_{u,p}}e^{-\bar{\jmath}2\pi i\Omega_{u,p}}$.

The ranges of parameters $\Omega_{u,p}$ and $\tau_{u,p}$ are $\mathcal{R}_{\mathrm{an}} = [-1/2, 1/2)$ and $\mathcal{R}_{\mathrm{de}} = [0, 1/\Delta f)$, which can be uniformly divided into $N_{\mathrm{an}} = F_{\mathrm{an}}M$ and $N_{\mathrm{de}} = F_{\mathrm{de}}N_{\mathrm{v}}$ intervals, respectively, and F_{an} and F_{de} are fine factors. Approximating the true parameters located in a interval with the left hand side endpoint, the sampled steering vectors can be defined as $\mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(n_{\mathrm{an}}) \triangleq \mathbf{v}_{\mathrm{s}}(\frac{n_{\mathrm{an}}}{N_{\mathrm{an}}} - \frac{1}{2})$, $\mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(n_{\mathrm{de}}) \triangleq \mathbf{v}_{\mathrm{f}}(\frac{n_{\mathrm{de}}}{\Delta f N_{\mathrm{de}}})$. In this way, the beam matrices can be formed by $\mathbf{V}_{\mathrm{s}} \triangleq [\mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(0), \ldots, \mathbf{v}_{\mathrm{s}}^{\mathrm{s}}(N_{\mathrm{an}}-1)]$ and $\mathbf{V}_{\mathrm{f}} \triangleq [\mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(0), \ldots, \mathbf{v}_{\mathrm{f}}^{\mathrm{s}}(N_{\mathrm{de}}-1)]$. Then the SF domain channel can be approximated as [8]

$$\mathbf{H}_u^{\mathrm{SF}} \approx \mathbf{V}_{\mathrm{s}}\mathbf{H}_u\mathbf{V}_{\mathrm{f}}^{\mathrm{T}}, \quad (4)$$

where $\mathbf{H}_u \in \mathbb{C}^{N_{\mathrm{an}}\times N_{\mathrm{de}}}$ is the beam domain channel and is defined as $[\mathbf{H}_u]_{n_{\mathrm{an}},n_{\mathrm{de}}} \triangleq \sum_{p=0}^{P_u^{n_{\mathrm{an}},n_{\mathrm{de}}}} \gamma_{u,p}$, where $P_u^{n_{\mathrm{an}},n_{\mathrm{de}}}$ is the number of the paths whose directional cosines $\Omega_{u,p}$ and delays $\tau_{u,p}$ are in intervals $[\frac{n_{\mathrm{an}}}{N_{\mathrm{an}}} - \frac{1}{2}, \frac{n_{\mathrm{an}}+1}{N_{\mathrm{an}}} - \frac{1}{2})$ and $[\frac{n_{\mathrm{de}}}{\Delta f N_{\mathrm{de}}}, \frac{n_{\mathrm{de}}+1}{\Delta f N_{\mathrm{de}}})$, respectively.

Since the delay spread is no greater than $N_{\mathrm{cp}}T_{\mathrm{s}}$ in the OFDM system, the non-zero elements of the beam domain channel are limited in the interval $[0, N_{\mathrm{f}})$ along the delay dimension, where $N_{\mathrm{f}} \triangleq \lceil\frac{N_{\mathrm{cp}}N_{\mathrm{de}}}{N_{\mathrm{sc}}}\rceil$. Besides, the beam matrices are all partial discrete Fourier transformation (DFT) matrices and can be rewritten as

$$\mathbf{V}_{\mathrm{s}} = \mathbf{I}_{M,N_{\mathrm{an}}}\mathbf{F}_{N_{\mathrm{an}}}\mathbf{\Pi}_{N_{\mathrm{an}}}^{N_{\mathrm{an}}/2}, \quad (5)$$

$$\mathbf{V}_{\mathrm{f}} = \mathbf{I}_{N_{\mathrm{v}},N_{\mathrm{de}}}\mathbf{F}_{N_{\mathrm{de}}}\mathbf{\Phi}_{N_{\mathrm{de}}}^{k_0}, \quad (6)$$

where $\mathbf{I}_{a,b} \in \mathbb{C}^{a\times b}$ is the matrix $[\mathbf{I}_{a,b}]_{n,n} = 1, n = 0, 1, \ldots, \min(a,b)-1$ and the other elements being 0, $\mathbf{F}_N \in \mathbb{C}^{N\times N}$ is the DFT matrix, $\mathbf{\Pi}_N^a \in \mathbb{C}^{N\times N}$ is the circular shift matrix satisfying $[\mathbf{\Pi}_N^a\mathbf{x}]_n = [\mathbf{x}]_{<n+a>_N}$ with $<\cdot>$ being modulo N, and $\mathbf{\Phi}_N^k \triangleq \mathrm{diag}([1, e^{\bar{\jmath}2\pi\frac{k}{N}}, \ldots, e^{\bar{\jmath}2\pi\frac{k(N-1)}{N}}])$.

C. Signal Model for Channel Acquisition

The comb-type pilot structure [9] over frequency dimension is widely used in 5G wireless communication for frequency multiplexing [10]. Using all subcarriers can be treated as a special case of comb-type pilot structure, and the comb-type pilot is compatible with any frame structure used in TDD uplink channel estimation. Therefore, the comb-type structure is considered in this paper.

For comb-type pilot structure, valid subcarriers are divided into N_{c} evenly spaced subcarrier groups, which is referred to as the combs. Each of combs has $N_{\mathrm{v}}^{\mathrm{p}} = N_{\mathrm{v}}/N_{\mathrm{c}}$ subcarriers to transmit pilots of $U_{\mathrm{c}} = U/N_{\mathrm{c}}$ UTs. Here for simplicity, N_{v} and U are assumed divisible by N_{c}. Considering UTs using the n_{c}-th comb only, let $\mathbf{x}_u$ denote the pilot sequences sent by UT u with length of $N_{\mathrm{v}}^{\mathrm{p}}$ over subcarriers used by the n_{c}-th comb. The received pilot signal at BS can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \sum_{u=0}^{U_{\mathrm{c}}-1}\mathbf{V}_{\mathrm{s}}\mathbf{H}_u(\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}})^{\mathrm{T}}\mathbf{X}_u + \mathbf{Z}_{n_{\mathrm{c}}} \in \mathbb{C}^{M\times N_{\mathrm{v}}^{\mathrm{p}}}, \quad (7)$$

where $\mathbf{X}_u = \mathrm{diag}(\mathbf{x}_u)$, $\mathbf{Z}_{n_{\mathrm{c}}}$ denotes the noise with distribution $\mathcal{CN}(\mathrm{vec}(\mathbf{Z}_{n_{\mathrm{c}}}); 0, \sigma_z^2\mathbf{I}_{MN_{\mathrm{v}}^{\mathrm{p}}})$, $\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}} = \mathbf{S}_{N_{\mathrm{v}}}^{N_{\mathrm{c}},n_{\mathrm{c}}}\mathbf{V}_{\mathrm{f}} \in \mathbb{C}^{N_{\mathrm{v}}^{\mathrm{p}}\times N_{\mathrm{de}}}$ is the modified frequency beam matrix for the n_{c}-th comb and $\mathbf{S}_N^{a,b} \in \mathbb{C}^{\frac{N}{a}\times N}$ is a selection matrix satisfying $[\mathbf{S}_N^{a,b}]_{i,ai+b} = 1, i = 0, 1, \ldots, N/a-1$ and the other elements being 0. Therefore, with (6), $\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}}$ can be rewritten as

$$\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}} = \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}^{N_{\mathrm{c}}}\mathbf{\Phi}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}}, \quad (8)$$

where $\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}^{N_{\mathrm{c}}} \triangleq [\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}, \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}, \ldots, \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}]$ with $N_{\mathrm{de}}^{\mathrm{p}} = F_{\mathrm{de}}N_{\mathrm{v}}^{\mathrm{p}}$, and $\mathbf{\Phi}_{N_{\mathrm{de}}}^{n_{\mathrm{c}}} = \mathrm{diag}([1, e^{\bar{\jmath}2\pi\frac{n_{\mathrm{c}}}{N_{\mathrm{de}}}}, \ldots, e^{\bar{\jmath}2\pi\frac{n_{\mathrm{c}}(N_{\mathrm{de}}-1)}{N_{\mathrm{de}}}}])$. Combined with the fact that non-zero elements of the beam domain channel are limited in the interval $[0, N_{\mathrm{f}})$ along the delay dimension, beam domain channel matrix and frequency beam matrix can be expressed by an equivalent form with a reduced size as

$$\widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}} \triangleq \mathbf{H}_u\mathbf{\Phi}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}}\mathbf{I}_{N_{\mathrm{de}},N_{\mathrm{de}}^{\mathrm{p}}} \in \mathbb{C}^{N_{\mathrm{an}}\times N_{\mathrm{de}}^{\mathrm{p}}}, \quad (9)$$

$$\widetilde{\mathbf{V}}_{\mathrm{f,p}} \triangleq \mathbf{I}_{N_{\mathrm{v}}^{\mathrm{p}},N_{\mathrm{de}}^{\mathrm{p}}}\mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}} \in \mathbb{C}^{N_{\mathrm{v}}^{\mathrm{p}}\times N_{\mathrm{de}}^{\mathrm{p}}}, \quad (10)$$

where the relation $\mathbf{H}_u(\mathbf{V}_{\mathrm{f,p}}^{n_{\mathrm{c}}})^{\mathrm{T}} = \widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}}(\widetilde{\mathbf{V}}_{\mathrm{f,p}})^{\mathrm{T}}$ holds when $N_{\mathrm{f}} \le N_{\mathrm{de}}^{\mathrm{p}}$. With (9) and (10), the received pilot signal can be rewritten as

$$\begin{aligned}\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} &= \sum_{u=0}^{U_{\mathrm{c}}-1}\mathbf{V}_{\mathrm{s}}\widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}}\widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}}\mathbf{X}_u + \mathbf{Z}_{n_{\mathrm{c}}} \\ &= \mathbf{V}_{\mathrm{s}}\widetilde{\mathbf{H}}_{n_{\mathrm{c}}}\mathbf{X}^{\mathrm{T}} + \mathbf{Z}_{n_{\mathrm{c}}},\end{aligned} \quad (11)$$

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where $\mathbf{X} = [\mathbf{X}_0\widetilde{\mathbf{V}}_{\mathrm{f,p}}, \mathbf{X}_1\widetilde{\mathbf{V}}_{\mathrm{f,p}}, \ldots, \mathbf{X}_{U_{\mathrm{c}}-1}\widetilde{\mathbf{V}}_{\mathrm{f,p}}]$ and $\widetilde{\mathbf{H}}_{n_{\mathrm{c}}} = [\widetilde{\mathbf{H}}_0^{n_{\mathrm{c}}}, \widetilde{\mathbf{H}}_1^{n_{\mathrm{c}}}, \ldots, \widetilde{\mathbf{H}}_{U_{\mathrm{c}}-1}^{n_{\mathrm{c}}}]$. With vectorization, the received pilot signal can be rewritten as

$$\mathbf{y}_{n_{\mathrm{c}}} = \mathbf{A}\mathbf{h}_{n_{\mathrm{c}}} + \mathbf{z}_{n_{\mathrm{c}}}, \tag{12}$$

where $\mathbf{A} \triangleq \mathbf{X}^{\mathrm{T}} \otimes \mathbf{V}_{\mathrm{s}} \in \mathbb{C}^{MN_{\mathrm{v}}^{\mathrm{p}} \times N_{\mathrm{an}}N_{\mathrm{de}}^{\mathrm{p}}}$, $\mathbf{y}_{n_{\mathrm{c}}}$, $\mathbf{h}_{n_{\mathrm{c}}}$ and $\mathbf{z}_{n_{\mathrm{c}}}$ are vectorizations of $\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}}$, $\widetilde{\mathbf{H}}_{n_{\mathrm{c}}}$ and $\mathbf{Z}_{n_{\mathrm{c}}}$, respectively. With the signal model in (12), the beam domain channel can be acquired by numerous channel estimation methods for linear models, such as the GGAMP-SBL algorithm.

III. Channel Estimation With Generalized Phase Shift Pilots

A. GPSP

Before presenting GPSPs, phase shift pilots (PSPs), which are a class of widely used pilots, should be introduced first. A phase shift pilot with length N can be expressed as

$$\mathbf{x}_u^{\mathrm{ps}} = \mathbf{x}_{\mathrm{r}} \odot \mathbf{f}_{\phi_u}, \tag{13}$$

where $\mathbf{x}_{\mathrm{r}}$ is the root sequence with length N shared by all users, $\mathbf{f}_{\phi_u} = [\mathbf{F}_N]_{:,\phi_u}$ and ϕ_u is the phase shift factor of the u-th UT. Utilizing the fact that $\mathbf{F}_N\mathbf{D}_{\phi_u} = \mathbf{\Pi}_N^{\phi_u}\mathbf{F}_N$, where $\mathbf{D}_{\phi_u} = \mathrm{diag}(\mathbf{f}_{\phi_u})$, PSPs allow channel matrices to delicately shift along the delay dimension to achieve optimal estimation performance [5], [11]. Then, GPSPs is defined as

$$\mathbf{x}_u \triangleq \mathbf{X}_{\mathrm{r}}\mathbf{\Phi}_{\mathrm{ps}}\mathbf{w}_u, \tag{14}$$

where $\mathbf{X}_{\mathrm{r}} = \mathrm{diag}(\mathbf{x}_{\mathrm{r}})$, $\mathbf{\Phi}_{\mathrm{ps}} = [\mathbf{f}_{\phi_0}, \mathbf{f}_{\phi_1} \ldots, \mathbf{f}_{\phi_{m-1}}]$, $\mathbf{w}_u$ is the $m \times 1$ vector composed of the weight sequence for the u-th UT and m is the number of used phase shift factors. Thus, GPSPs are linear combinations of multiple PSPs with different weights, and can be rewritten as

$$\mathbf{x}_u \triangleq \mathbf{x}_{\mathrm{r}} \odot \mathbf{x}_u^{\mathrm{ps}}, \tag{15}$$

where $\mathbf{x}_u^{\mathrm{ps}} = \sum_{k=0}^{m-1}[\mathbf{w}_u]_k\mathbf{f}_{\phi_k}$. Note that GPSPs are consistent with the definition of PSPs in (13) while the phase shift sequences $\mathbf{x}_u^{\mathrm{ps}}$ can be generated from an m-dimensional subspace formed by the columns of $\mathbf{\Phi}_{\mathrm{ps}}$. Therefore, GPSPs universally generalize the class of PSPs. Specially, when $\mathbf{\Phi}_{\mathrm{ps}} = \mathbf{F}_N$ and $\mathbf{w}_u$ is a vector with one element being 1 and the others being 0, the GPSPs are reduced to APSPs. Considering the distinct properties of ZC sequences [12], [13], we use the Zadoff-Chu (ZC) sequence [14] for the root sequence $\mathbf{x}_{\mathrm{r}}$, i.e.

$$[\mathbf{x}_N^r]_n \triangleq e^{-\bar{\jmath}\pi\frac{rn(n+\langle N\rangle_2)}{N}},\ n = 0, 1, 2, \ldots, N-1, \tag{16}$$

where the root r is an integer relatively prime to N. Then, we have the following theorem.

Theorem 1: When the phase shifts are chosen uniformly and the weight sequence $\mathbf{w}_u$ is a ZC sequence of length m, GPSP has constant modulus with a modulus of $\sqrt{m}$.

Proof: With the phase shifts chosen uniformly, $\mathbf{\Phi}_{\mathrm{mp}}$ can be expressed as $[\mathbf{f}_{\phi_0}, \mathbf{f}_{\phi_0+s} \ldots, \mathbf{f}_{\phi_0+N-s}]$, where $s = N/m$ is the spacing length of two adjacent phase shifts. Then we have

$$\mathbf{x}_u = \mathbf{X}_{\mathrm{r}}\mathbf{D}_{\phi_0}\begin{bmatrix}\mathbf{F}_m\mathbf{w}_u\\ \mathbf{F}_m\mathbf{w}_u\\ \vdots\\ \mathbf{F}_m\mathbf{w}_u\end{bmatrix} = \sqrt{m}\mathbf{X}_{\mathrm{r}}\mathbf{D}_{\phi_0}\begin{bmatrix}\tilde{\mathbf{w}}_u\\ \tilde{\mathbf{w}}_u\\ \vdots\\ \tilde{\mathbf{w}}_u\end{bmatrix}, \tag{17}$$

where $\tilde{\mathbf{w}}_u = \mathbf{F}_m\mathbf{w}_u$ and $\mathbf{D}_{\phi_0} = \mathrm{diag}(\mathbf{f}_{\phi_0})$. When $\mathbf{w}_u$ is a ZC sequence of length m, $\tilde{\mathbf{w}}_u$ is still a ZC sequence of length m [12], [13]. $\mathbf{X}_{\mathrm{r}}$ and $\mathbf{D}_{\phi_0}$ are diagonal matrices with unit modulus diagonal elements, so the modulus of $\mathbf{x}_u$ is constant. This completes the proof. ∎

In the rest of the paper, the GPSPs are assumed to satisfy the conditions of Theorem 1. For simplicity, ϕ_0 is set to 0. In this case, $s = N/m$ is referred to as the spacing length of the fixed phase shifts and the root of $\mathbf{w}_u$ is denoted by q. In an OFDM system, like the phase shift pilots, GPSPs are applied in the frequency domain and transformed to the time domain by IDFT. Denote $\tilde{\mathbf{x}}_u = \frac{1}{\sqrt{N}}\mathbf{F}_N^{\mathrm{H}}\mathbf{x}_u$, $\tilde{\mathbf{x}}_{\mathrm{r}} = \frac{1}{\sqrt{N}}\mathbf{F}_N^{\mathrm{H}}\mathbf{x}_{\mathrm{r}}$, and when $\mathbf{x}_{\mathrm{r}}$ is a ZC sequence, $\tilde{\mathbf{x}}_{\mathrm{r}}$ is also a ZC sequence with a new root $\tilde{r}$ [12], [13]. For the IDFT of GPSPs, we have the following theorem.

Theorem 2: When the root sequence $\mathbf{x}_{\mathrm{r}}$ is a ZC sequence and $q + s\tilde{r}$ is relative prime to m, the IDFT of $\mathbf{x}_u$ has constant modulus with a modulus of $\sqrt{m}$.

Proof: Note that the GPSP $\mathbf{x}_u$ can be rewritten as $\sum_{k=0}^{m-1}[\mathbf{w}_u]_k\mathbf{D}_{\phi_k}\mathbf{x}_{\mathrm{r}}$. Then we have

$$\tilde{\mathbf{x}}_u = \sum_{k=0}^{m-1}[\mathbf{w}_u]_k\mathbf{\Pi}_N^{-\phi_k}\tilde{\mathbf{x}}_{\mathrm{r}}. \tag{18}$$

Therefore, we have

$$[\tilde{\mathbf{x}}_u]_n^{\mathrm{H}}[\tilde{\mathbf{x}}_u]_n = \sum_{k=0}^{m-1}\sum_{l=0}^{m-1}[\mathbf{w}_u]_k^{\mathrm{H}}[\tilde{\mathbf{x}}_{\mathrm{r}}]_{n-\phi_k}^{\mathrm{H}}[\mathbf{w}_u]_l[\tilde{\mathbf{x}}_{\mathrm{r}}]_{n-\phi_l}. \tag{19}$$

The periodicity of ZC sequences allows the change of variable $l \to l + k$, and with $\phi_0 = 0$, we have that $\phi_k = ks$. Thus, for the case that N and m are both even, we have

$$[\tilde{\mathbf{x}}_u]_n^{\mathrm{H}}[\tilde{\mathbf{x}}_u]_n = \sum_{k=0}^{m-1}e^{-\bar{\jmath}\pi q\frac{k^2}{m}}e^{-\bar{\jmath}\pi\tilde{r}\frac{sk^2-2nk}{m}}\sum_{l=0}^{m-1}e^{-\bar{\jmath}2\pi(q+s\tilde{r})\frac{kl}{m}}. \tag{20}$$

Since $q + s\tilde{r}$ is relatively prime to m, $< (q + s\tilde{r})k >_m = 0$ if and only if k is 0 for $k \in [0, 1, \ldots, m-1]$. Thus, the second summation in (20) is 0 unless $k = 0$. Then it can be easily shown that $|[\tilde{\mathbf{x}}_u]_n| = \sqrt{m}$ for all n. In this way, the proof can be completed for the case with N and m both even. The proof for odd N or m is similar. ∎

From Theorem 2, when the GPSPs are used in the frequency domain, the low peak to average power ratio (PAPR) is guaranteed directly in the time domain, which allows higher average transmit power and then facilitates the channel estimation [15]. Therefore, the root sequence of GPSPs is set to a ZC sequence and $q + s\tilde{r}$ is assumed prime to m in subsequent discussions.

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Furthermore, as for the correlation property of the IDFT of GPSPs, we have following theorem.

Theorem 3: The autocorrelation function of the IDFT of GPSPs is

$$R_a[p] = \tilde{\mathbf{x}}_u^{\mathrm{H}} \mathbf{\Pi}_N^p \tilde{\mathbf{x}}_u = mN\delta[p], \tag{21}$$

and the modulus of crosscorrelation function is

$$|R_c^{q,q'}[p]| = \sum_{k=0}^{m-1} \sqrt{m} N \delta[p - ks]. \tag{22}$$

Proof: Substituting(18) into (21), we have

$$R_a[p] = \sum_{k=0}^{m-1} \sum_{l=0}^{m-1} [\mathbf{w}_u]_k^{\mathrm{H}} [\mathbf{w}_u]_l \tilde{\mathbf{x}}_{\mathrm{r}}^{\mathrm{H}} \mathbf{\Pi}_N^{\phi_k - \phi_l + p} \tilde{\mathbf{x}}_{\mathrm{r}}. \tag{23}$$

When $< p >_s \neq 0$, $< \phi_k - \phi_l + p >_N \neq 0$. Then, utilizing the autocorrelation property of a ZC sequence, $\tilde{\mathbf{x}}_{\mathrm{r}}^{\mathrm{H}} \mathbf{\Pi}_N^{\phi_k - \phi_l + p} \tilde{\mathbf{x}}_{\mathrm{r}} = 0$ and thus $R_a[p] = 0$. When $< p >_s = 0$, there exit k, l such that $< \phi_k - \phi_l + p >_N = 0$. In this case, we have $R_a[p] = N \sum_{k=0}^{m-1} [\mathbf{w}_u]_k^{\mathrm{H}} [\mathbf{w}_u]_{<k+\frac{p}{s}>_m} = N \mathbf{w}_u^{\mathrm{H}} \mathbf{\Pi}_N^{\frac{p}{s}} \mathbf{w}_u$, which equals to m only when $p = 0$ and otherwise equals to 0. In this way, the part of autocorrelation has been proved.

Substituting (18) into (22), we have

$$R_c^{u,u'}[p] = \sum_{k=0}^{m-1} \sum_{l=0}^{m-1} [\mathbf{w}_{u'}]_k^{\mathrm{H}} [\mathbf{w}_u]_l \tilde{\mathbf{x}}_{\mathrm{r}}^{\mathrm{H}} \mathbf{\Pi}_N^{\phi_k - \phi_l + p} \tilde{\mathbf{x}}_{\mathrm{r}}. \tag{24}$$

Similarly, when $< p >_s \neq 0$, $R_c[p] = 0$, and when $< p >_s = 0$, $R_c^{u,u'}[p] = N \mathbf{w}_{u'}^{\mathrm{H}} \mathbf{\Pi}_N^{\frac{p}{s}} \mathbf{w}_u$. Then utilizing the crosscorrelation property of ZC sequences, we have $|R_c^{u,u'}[p]| = \sqrt{m} N$ when $< p >_s = 0$. This completes the proof. ∎

From Theorem 3, the IDFT of GPSPs has the optimal autocorrelation property, and the modulus of elements in crosscorrelation function does not exceed $1/\sqrt{m}$ of the peak of autocorrelation function. Therefore, when m is large enough, the crosscorrelation between different GPSPs is negligible, which enables GPSPs to behave similarly to orthogonal pilots and then benefits the channel estimation. Then, according to the theorems, GPSPs can be specified as below.

$$\mathbf{x}_u = \mathbf{x}_{N,m}^{r,q_u} = \mathbf{x}_N^r \odot \mathbf{x}_{m,N}^{q_u}, \tag{25}$$

where $[\mathbf{x}_{m,N}^{q_u}]_n \triangleq [\mathbf{x}_m^{q_u}]_{\langle n \rangle_m}, n = 1, 2, \ldots, N-1$ with N specified as $N_{\mathrm{v}}^{\mathrm{p}}$, and q_u is the root for ZC sequence $\mathbf{x}_m^{q_u}$. In this case, the parameter s represents the number of periodic extension of $\mathbf{x}_m^{q_u}$ in the $\mathbf{x}_{m,N}^{q_u}$.

B. Inter-User Interference Analysis

Considering the channel estimation in a single cell, $\mathbf{x}_{N_{\mathrm{v}}^{\mathrm{p}}}^r$ is the same for all users, while $\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u}$ changes for different users in the same comb and is reused for users in different combs. With the GPSPs, the received signal model is

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} = \sum_{u=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}} \widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}} \widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}} \mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r \mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u} + \mathbf{Z}_{n_{\mathrm{c}}}, \tag{26}$$

where $\mathbf{X}_u$ in (7) is substituted by $\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r \mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u}$ with $\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r = \mathrm{diag}(\mathbf{x}_{N_{\mathrm{v}}^{\mathrm{p}}}^r)$ and $\mathbf{X}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u} = \mathrm{diag}(\mathbf{x}_{m,N_{\mathrm{v}}^{\mathrm{p}}}^{q_u})$. We post-multiply the received signal by the conjugate transpose of the pilot matrix, which can be expressed as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} \mathbf{X}_u^{\mathrm{H}} = \sum_{u'=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}} \widetilde{\mathbf{H}}_{u'}^{n_{\mathrm{c}}} \widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}} \mathbf{X}_{u'} \mathbf{X}_u^{\mathrm{H}} + \bar{\mathbf{Z}}_{n_{\mathrm{c}}}, \tag{27}$$

where $\bar{\mathbf{Z}}_{n_{\mathrm{c}}} \triangleq \mathbf{Z}_{n_{\mathrm{c}}} \mathbf{X}_u^{\mathrm{H}}$. Note that $\mathbf{X}_u \widetilde{\mathbf{V}}_{\mathrm{f,p}}$ can be rewritten as

$$\mathbf{X}_u \widetilde{\mathbf{V}}_{\mathrm{f,p}} = \mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r \widetilde{\mathbf{V}}_{\mathrm{f,p}} \mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u}), \tag{28}$$

where $\mathrm{circ}(\mathbf{x})$ denotes the circulant matrix with the first column being $\mathbf{x}$, and $\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u} = \frac{1}{N_{\mathrm{de}}^{\mathrm{p}}} \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}}^{\mathrm{H}} \mathbf{x}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u}$. From (28), $\mathbf{X}_u$ can be split into $\mathbf{X}_{N_{\mathrm{v}}^{\mathrm{p}}}^r$ in the frequency domain and $\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u})$ in the frequency beam domain, respectively. Substituting (28) into (27), we have

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} \mathbf{X}_u^{\mathrm{H}} = \sum_{u'=0}^{U_{\mathrm{c}}-1} \mathbf{V}_{\mathrm{s}} \widetilde{\mathbf{H}}_{u'}^{n_{\mathrm{c}}} (\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u,q_{u'}}))^{\mathrm{T}} \widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}} + \bar{\mathbf{Z}}_{n_{\mathrm{c}}}, \tag{29}$$

where $\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u,q_{u'}}) = (\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u}))^{\mathrm{H}} \mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_{u'}})$ as the product of two circulant matrices is also a circulant matrix and $[\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u,q_{u'}}]_n = (\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u})^{\mathrm{H}} \mathbf{\Pi}_{N_{\mathrm{de}}^{\mathrm{p}}}^n \widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_{u'}}$. Then utilizing Theorem 3, (29) can be rewritten as

$$\mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} \mathbf{X}_u^{\mathrm{H}} = \mathbf{V}_{\mathrm{s}} (\widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}} + \mathbf{I}_u^{\mathrm{B}}) \widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}} + \bar{\mathbf{Z}}_{n_{\mathrm{c}}}, \tag{30}$$

where $\mathbf{I}_u^{\mathrm{B}} = \sum_{u' \neq u} \widetilde{\mathbf{H}}_{u'}^{n_{\mathrm{c}}} (\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u,q_{u'}}))^{\mathrm{T}}$ is the interference of UT u in the beam domain. Moreover, according to Theorem 3, the modulus of elements in $(\mathrm{circ}(\widetilde{\mathbf{x}}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u,q_{u'}}))^{\mathrm{T}}$ are no more than $\frac{1}{\sqrt{m}}$ when $q_{u'} \neq q_u$. Then we have the following theorem.

Theorem 4: When $m \to \infty$ and the number of propagation paths is finite, the inter-user interference caused by GPSPs disappears, i.e.

$$\lim_{m \to \infty} [\mathbf{I}_{n_{\mathrm{c}},u}^{\mathrm{B}}]_{n_{\mathrm{an}},n_{\mathrm{de}}} = 0, \tag{31}$$

for arbitrary n_{an} and n_{de}, and

$$\lim_{m \to \infty} \mathbf{Y}_{n_{\mathrm{c}}}^{\mathrm{SF}} \mathbf{X}_u^{\mathrm{H}} = \mathbf{V}_{\mathrm{s}} \widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}} \widetilde{\mathbf{V}}_{\mathrm{f,p}}^{\mathrm{T}} + \bar{\mathbf{Z}}_{n_{\mathrm{c}}}. \tag{32}$$

Proof: Proof is omitted due to the space limitation. ∎

Theorem 4 suggests that when the length of weight sequences is sufficiently long, the inter-user interference can be negligible, thanks to the crosscorrelation properties of GPSPs. Therefore, Theorem 4 indicates the feasibility of the GPSPs in channel estimation without statistical CSI since it does not depend on any pilot scheduling algorithms.

C. Efficient Channel Estimation With GPSPs

With GPSPs, the beam domain channel can be obtained by GGAMP-SBL algorithm [7], and then the SF domain channel can be recovered according to (4) and $\mathbf{H}_u = \widetilde{\mathbf{H}}_u^{n_{\mathrm{c}}} \mathbf{I}_{N_{\mathrm{de}}^{\mathrm{p}},N_{\mathrm{de}}} (\mathbf{\Phi}_{N_{\mathrm{de}}}^{k_{n_{\mathrm{c}}}})^{\mathrm{H}}$, which can be easily obtained from (9). When the GGAMP-SBL algorithm is applied, the computational complexity mainly concentrates on calculating $\mathbf{A}\hat{\mathbf{h}}$ and $\mathbf{A}^{\mathrm{H}}\mathbf{y}$, where $\hat{\mathbf{h}}$ is the vector with the same size of $\mathbf{h}_{n_{\mathrm{c}}}$ and $\mathbf{y}$

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is the vector with the same size of $\mathbf{y}_{n_c}$. From (12), $\mathbf{A}\hat{\mathbf{h}}$ and $\mathbf{A}^{\mathrm{H}}\mathbf{y}$ can be calculated as

$$\mathbf{A}\hat{\mathbf{h}} = \sum_{u=0}^{U_c-1} \mathbf{V}_s \hat{\mathbf{H}}_u (\mathbf{X}_u \widetilde{\mathbf{V}}_{f,p})^{\mathrm{T}}, \tag{33}$$

$$\mathbf{A}^{\mathrm{H}}\mathbf{y} = \mathrm{vec}([\hat{\mathbf{H}}_0, \hat{\mathbf{H}}_1, \ldots, \hat{\mathbf{H}}_{U_c-1}]), \tag{34}$$

where $\mathrm{vec}(\mathbf{Y}) = \mathbf{y}$ and $\hat{\mathbf{H}}_u = \mathbf{V}_s^{\mathrm{H}} \mathbf{Y} (\mathbf{X}_u \widetilde{\mathbf{V}}_{f,p})^*$. Since the beam matrices are all partial DFT matrices, these matrix multiplications can be implemented by FFT and inverse FFT (IFFT). The numbers of complex multiplications of directly calculating $\mathbf{A}\hat{\mathbf{h}}$ and $\mathbf{A}^{\mathrm{H}}\mathbf{y}$ based on (33) and (34) are both

$$N_{\mathrm{di}} = M_{\mathrm{all}}^{\mathrm{p}} U_c + \frac{1}{2} N_{\mathrm{all}}^{\mathrm{p}} U_c \log_2(\frac{N_{\mathrm{all}}^{\mathrm{p}}}{4}), \tag{35}$$

where $M_{\mathrm{all}}^{\mathrm{p}} = M N_v^{\mathrm{p}}$, $N_{\mathrm{all}}^{\mathrm{p}} = N_{\mathrm{an}} N_{\mathrm{de}}^{\mathrm{p}}$ and the number of complex multiplications in FFT is measured as $N/2\log(N/4)$ for N-point FFT since no multiplication is needed when FFT size is not greater than 4.

To further reduce the computational complexity, the periodical part in GPSPs can be utilized to decompose the original large-size FFT into several small-size FFT. The efficient implementation of computing $\mathbf{A}\hat{\mathbf{h}}$ can be divided into two stages. The first stage is computed for different users as

$$\hat{\mathbf{H}}_u^1 = \hat{\mathbf{H}}_u (\mathbf{X}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u} \mathbf{D}_{\mathrm{F}} \mathbf{P}_{m,F_{\mathrm{de}}s})^{\mathrm{T}}, \tag{36}$$

where $\hat{\mathbf{H}}_u^1$ is the output of the first-stage computation, $\mathbf{X}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u} = \mathrm{diag}(\mathbf{x}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u})$, $\mathbf{D}_{\mathrm{F}} = \mathrm{Bdiag}(\mathbf{F}_m, \mathbf{F}_m, \ldots, \mathbf{F}_m) \in \mathbb{C}^{N_{\mathrm{de}}^{\mathrm{p}} \times N_{\mathrm{de}}^{\mathrm{p}}}$ with $\mathrm{Bdiag}(\mathbf{A}_0, \mathbf{A}_1, \ldots)$ denoting a block diagonal matrix with the elements $\mathbf{A}_i$ located along the main diagonal, and $\mathbf{P}_{m,F_{\mathrm{de}}s}$ is the permutation matrix constructed by letting $[\mathbf{P}_{m,F_{\mathrm{de}}s}]_{k,F_{\mathrm{de}}sl_1+l_2} = 1$, $k = 0, 1, \ldots, N_{\mathrm{de}}^{\mathrm{p}} - 1$, $l_1 = \langle k \rangle_m$, $l_2 = \lfloor k/m \rfloor$ and other elements be 0. Then $\hat{\mathbf{H}}_u^1$ for different users are summed to compute the second-stage efficient computation, which can be expressed as

$$\mathbf{A}\hat{\mathbf{h}} = \mathbf{V}_s (\sum_{u=0}^{U_c-1} \hat{\mathbf{H}}_u^1)(\mathbf{V}_{\mathrm{f}}^{\mathrm{ef}})^{\mathrm{T}}, \tag{37}$$

where $\mathbf{V}_{\mathrm{f}}^{\mathrm{ef}} \triangleq \mathbf{X}_{N_v^{\mathrm{p}}}^{r} \mathbf{I}_{N_v^{\mathrm{p}}, N_{\mathrm{de}}^{\mathrm{p}}} \mathbf{P}_{F_{\mathrm{de}}s,m}^{\mathrm{T}} \mathbf{D}_{s,m} \mathbf{P}_{F_{\mathrm{de}}s,m}$, $\mathbf{P}_{F_{\mathrm{de}}s,m}$ is another permutation matrix, and $\mathbf{D}_{s,m}$ can be defined as

$$\begin{aligned} \mathbf{D}_{s,m} &\triangleq \mathbf{P}_{F_{\mathrm{de}}s,m} \mathbf{F}_{N_{\mathrm{de}}^{\mathrm{p}}} \mathbf{P}_{m,F_{\mathrm{de}}s}^{\mathrm{T}} \mathbf{D}_{\mathrm{F}}^{\mathrm{H}} \mathbf{P}_{F_{\mathrm{de}}s,m}^{\mathrm{T}} \\ &= \mathrm{Bdiag}\left(\mathbf{F}_{F_{\mathrm{de}}s}, \mathbf{F}_{F_{\mathrm{de}}s} \widetilde{\mathbf{\Phi}}_{N_{\mathrm{de}}^{\mathrm{p}}}^{1}, \ldots, \mathbf{F}_{F_{\mathrm{de}}s} \widetilde{\mathbf{\Phi}}_{N_{\mathrm{de}}^{\mathrm{p}}}^{m-1}\right), \end{aligned} \tag{38}$$

where $\widetilde{\mathbf{\Phi}}_{N_{\mathrm{de}}^{\mathrm{p}}}^{k} \triangleq \mathbf{I}_{F_{\mathrm{de}}s, N_{\mathrm{de}}^{\mathrm{p}}} \mathbf{\Phi}_{N_{\mathrm{de}}^{\mathrm{p}}}^{k}$. Similarly, the efficient computation of $\mathbf{A}^{\mathrm{H}}\mathbf{y}$ can be divided into two stages as

$$\mathbf{Y}^1 = \mathbf{V}_s^{\mathrm{H}} \mathbf{Y} (\mathbf{V}_{\mathrm{f}}^{\mathrm{ef}})^*, \tag{39}$$

$$\hat{\mathbf{H}}_u = \mathbf{Y}^1 (\mathbf{X}_{m,N_{\mathrm{de}}^{\mathrm{p}}}^{q_u} \mathbf{D}_{\mathrm{F}} \mathbf{P}_{m,F_{\mathrm{de}}s})^*. \tag{40}$$

In this way, the original $N_{\mathrm{de}}^{\mathrm{p}}$-dimensional FFT performed U_c times in the direct implementation is replaced by m-dimensional FFT performed $U_c F_{\mathrm{de}} s$ times and $F_{\mathrm{de}}s$-dimensional FFT performed m times in the efficient implementation, by which the computational complexity is reduced. Considering permutation matrix is implemented by reindexing, the number of complex multiplications of $\mathbf{A}\hat{\mathbf{h}}$ or $\mathbf{A}^{\mathrm{H}}\mathbf{y}$ based on the efficient implementation is

$$\begin{aligned} N_{\mathrm{ef}} = M_{\mathrm{all}}^{\mathrm{p}} &+ \frac{1}{2} N_{\mathrm{all}}^{\mathrm{p}} \log_2(\frac{N_{\mathrm{an}} F_{\mathrm{de}} s}{16}) \\ &+ \frac{1}{2} N_{\mathrm{all}}^{\mathrm{p}} U_c \log_2\left(\frac{m}{4}\right) + N_{\mathrm{all}}^{\mathrm{p}} (U_c + 1). \end{aligned} \tag{41}$$

Since the computational complexity of the GGAMP-SBL algorithm is dominated by calculating $\mathbf{A}\hat{\mathbf{h}}$ and $\mathbf{A}^{\mathrm{H}}\mathbf{y}$, $2N_{\mathrm{di}}$ and $2N_{\mathrm{ef}}$ can be regarded as an approximation of the complex multiplication number in one iteration of the GGAMP-SBL algorithm. With (35) and (41), the ratio between the numbers of complex multiplications of the efficient implementation and that of direct implementation can be expressed as

$$\frac{N_{\mathrm{ef}}}{N_{\mathrm{di}}} \approx \frac{1}{U_c} + \frac{U_c - 1}{U_c} \frac{2 + \log_2(\frac{m}{4})}{\log_2(\frac{N_{\mathrm{all}}^{\mathrm{p}}}{4})}. \tag{42}$$

Since m is much smaller than $N_{\mathrm{all}}^{\mathrm{p}}$, the number of complex multiplications of the proposed efficient implementation is significantly decreased. Moreover, when N_v^{p} is fixed, the smaller m leads to less needed complex multiplications. Specially, when $M \to \infty$, $N_{\mathrm{all}}^{\mathrm{p}} \to \infty$, we have that $N_{\mathrm{ef}}/N_{\mathrm{di}} \to 1/U_c$, which implies that the efficient implementation can reduce the number of complex multiplications to $1/U_c$ at most.

Combined with Theorem 4, the GPSPs provide a trade-off between performance and computational complexity. For a fixed N_v^{p}, the performance and computational complexity can be flexibly adjusted by m or s. For example, with smaller s, the interference becomes less while the complexity becomes higher due to larger m, and vice versa, according to (30) and (42), respectively.

TABLE I
PARAMETER SETTINGS

Parameter	Value
Number of BS antennas M	128
UT number U	48
Center frequency f_c	5.8GHz
Number of valid subcarriers N_v	1152
Number of subcarriers per comb N_v^{p}	96
Number of combs N_c	12
Subcarrier spacing Δf	15kHz
Number of subcarriers N_c	2048
CP length N_{cp}	144
Fine factor F_{an}, F_{de}	2

IV. Simulation Results

In this section, simulation results are provided to illustrate the performance of the proposed GPSPs for massive MIMO-OFDM channel acquisition. QuaDRiGa channel model [16] is adopted to generate channels with main parameters summarized in Table I, and the SF domain channel is acquired by the GGAMP-SBL algorithm. The normalized mean square

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error (NMSE) of the estimated SF domain channel is used as the performance metric. We compare the proposed GPSPs with the widely used ZC sequences defined in (16).

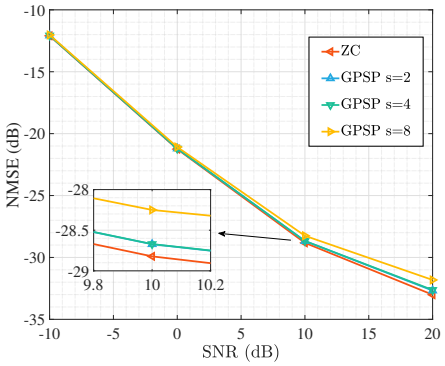


Fig. 1. Channel estimation performance with different pilots

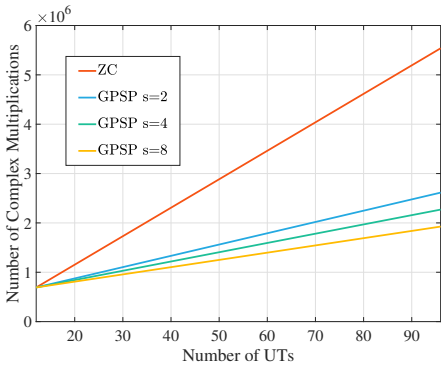


Fig. 2. The number of complex multiplications in one iteration under different pilots

Fig. 1 shows the NMSE performance of the channel estimation without the statistical CSI under different pilot configurations. When the number of subcarriers, $N_{\mathrm{v}}^{\mathrm{P}} = sm$, is fixed, GPSPs can achieve better performance with smaller s since larger m results in a corresponding reduction in the inter-user interference. When s is small enough, the performance of GPSPs aligns closely with that of the ZC sequences.

The comparison of the complex multiplication numbers required in one iteration for ZC sequences and GPSPs is illustrated in Fig. 2. The GGAMP-SBL algorithm for ZC sequences is implemented by (33) and (34), while it is implemented by (37) and (40) for GPSPs. Therefore, the numbers of complex multiplications of ZC sequence and GPSPs are in accordance with (35) and (41), respectively. For a fixed number of subcarriers $N_{\mathrm{v}}^{\mathrm{P}} = sm$, it is observed that an increase in the value of s results in a reduction in the number of complex multiplications.

The results in Fig. 1 and Fig. 2 collectively demonstrate the trade-off of GPSPs between performance and computational complexity. We can observe that, when GPSPs and ZC sequences exhibit close performance, the computational complexity of channel estimation with GPSPs can be reduced to the half of that with ZC sequences.

V. Conclusion

In this paper, we first formulated the estimation problem under the beam based channel model with the comb-type pilot structure. Then, we proposed the GPSPs and revealed their relevant properties. We analyzed the inter-user interference caused by GPSPs, which can be negligible when the length of weight sequence is large enough. Therefore, the use of the proposed GPSPs for channel acquisition is feasible. Then, with the property of GPSPs, we proposed an efficient implementation with much less complex multiplications for GGAMP-SBL channel estimation algorithm. Simulation results verified the superiority of the proposed GPSPs.

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